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RESEARCH

Multifaceted freshwater fish α-diversity exhibits discordant spatial patterns and distinct environmental associations across the United States

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Abstract Understanding how different dimensions of biodiversity vary across environmental gradients is important for characterizing large-scale patterns of freshwater biodiversity. Taxonomic diversity (TD), functional diversity (FD), and phylogenetic diversity (PD) describe different facets of biological communities and may exhibit either concordant or divergent spatial patterns. However, their relative spatial patterns and environmental associations in freshwater fish assemblages remain insufficiently understood at the continental scale. In this study, we integrated distribution data, functional traits, and phylogenetic information for 786 native freshwater fish species across 1873 watersheds in the contiguous United States to quantify TD, FD, and PD. Boosted Regression Tree models (BRTs) were used to assess the associations of these three diversity facets with contemporary climate, historical climatic change, and geographical factors. The three diversity facets exhibited pronounced spatial heterogeneity and partial spatial incongruence, with distinct patterns of association with environmental variables. TD exhibited a clear west–east gradient, with lower values in the western U.S. and higher values in the central and eastern regions. It was most strongly associated with precipitation during the warmest quarter, and the BRT model explained 86.9% of the observed variation. FD hotspots were concentrated along the Pacific and Atlantic coasts and in the Florida Peninsula, whereas most watersheds in the Central Plains showed relatively low FD. FD was most strongly associated with elevation and the change in mean annual temperature since the Last Glacial Maximum (LGMT), with the model explaining 65.5% of the variation. PD hotspots were distributed along the Atlantic Coast, the Florida Peninsula, and the Great Lakes region. Similar to FD, PD was most strongly associated with elevation and LGMT, with the model explaining 53.1% of the variation. Overall, the three diversity facets showed only partial spatial congruence and differed in their environmental associations, indicating that large-scale

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patterns of freshwater fish biodiversity are not fully captured by species richness alone. These findings provide complementary perspectives on freshwater fish biodiversity and may inform biodiversity assessment and conservation planning by considering taxonomic, functional, and phylogenetic dimensions alongside contemporary environmental gradients and historical climatic legacies.

Keywords Freshwater fish · Alpha diversity · Environmental factors · Taxonomic diversity · Functional diversity · Phylogenetic diversity

Introduction

Biodiversity conservation has become a critical priority in ecological research, especially in the context of accelerating human pressures leading to the rapid decline of various biological groups worldwide (Díaz et al. 2019; Su et al. 2021). Understanding the spatial distribution patterns of biodiversity and analyzing their driving mechanisms remain key objectives in biogeography and macroecology (Gaston 2000). Traditional biodiversity assessments primarily rely on taxonomic diversity (TD), particularly species richness, as the core indicator of α-diversity (Whittaker 1960). While taxonomic diversity is widely used across ecosystems (Swenson 2011; Kang et al. 2018; Chen et al. 2020), it has inherent limitations. Traditional taxonomic-diversity indices, such as total species count, the Menhinick index, the Margalef index, and Fisher's α index, quantify species abundance but fail to reflect functional traits or evolutionary relationships (Thukral 2017).

To address these shortcomings, functional diversity (FD) and phylogenetic diversity (PD) offer complementary perspectives, focusing on species' ecological roles and long-term evolutionary history. Functional diversity reveals species' ecological roles within ecosystems through key traits such as morphology, physiology, and resource use strategies, providing insights into ecosystem functioning (Jia et al. 2020; Liu et al. 2023). Among the multiple FD indices, functional richness is one of the core indicators, measuring the volume of functional space occupied by a biological community. Phylogenetic diversity is commonly measured by phylogenetic richness, which reflects the cumulative branch length of a species set on the phylogenetic tree (Faith 1992). TD, FD and PD indices are often closely correlated. For example, Tucker et al. (2018) demonstrated that the correlation between PD and FD strengthens as the number of functional traits included increases. However, this correlation weakens when trait evolutionary rates change or evolutionary optima shift. Thus, comparing TD, FD, and PD within a common analytical setting can help evaluate whether different biodiversity dimensions convey congruent or distinct information across spatial and evolutionary scales (Webb et al. 2002; Goswami et al. 2017).

Freshwater ecosystems occupy less than 1% of Earth's non-glacial surface but support about a quarter of the global vertebrate diversity (Allen and Pavelsky 2018). Freshwater fish not only contribute to food security and poverty alleviation through fisheries (Pelicice et al. 2023), but also play a crucial role in maintaining ecosystem functions, stability, and nutrient cycling (Vanni 2002; Villéger et al. 2017). However, freshwater biodiversity is under severe threat: since 1970, monitored freshwater-vertebrate populations have shown an average decline of 84% in abundance, a loss far greater than that observed in most other ecosystems (Watson et al. 2019; Lin et al. 2021). Therefore, in the context of rapid environmental change, it is useful to compare taxonomic, functional, and phylogenetic diversity indices and their environmental associations to inform freshwater-fish biodiversity assessment and conservation planning.

Among temperate regions worldwide, the U.S. harbors one of the richest freshwater fish faunas (Knouft and Page 2011), making it an ideal system for investigating large-scale biodiversity patterns. Previous studies have examined freshwater fish biodiversity of the U.S. from multiple perspectives, including taxonomic diversity, phylogenetic diversity, beta diversity, and responses to environmental change and human disturbance (Qian et al. 2021; Anas and Mandrak 2021; Xu et al. 2025). Although these studies have substantially advanced our understanding of freshwater fish biogeography, relatively few have systematically compared the spatial patterns and environmental associations of taxonomic, functional, and phylogenetic α-diversity across broad environmental gradients. Consequently, whether these three dimensions exhibit similar spatial patterns and environmental associations at the continental scale remains insufficiently understood.

Such comparisons are important because concordant patterns would suggest that different diversity dimensions capture broadly similar large-scale gradients, whereas divergent patterns would indicate that they provide complementary perspectives on the structure of freshwater fish assemblages. The contiguous U.S. is particularly well suited for such a comparison because of its pronounced gradients in climate, topography, and glacial history, which may differentially structure freshwater fish assemblages (Knouft and Page 2011).

To address these questions, this study compares the spatial patterns of taxonomic, functional, and phylogenetic α-diversity across 1,873 U.S. watersheds using native freshwater fish assemblages. We further evaluate their relationships with habitat characteristics, contemporary climate-energy conditions, and historical environmental factors to determine whether different biodiversity dimensions exhibit similar or distinct environmental associations at the continental scale. By examining multiple biodiversity dimensions within a unified spatial framework, this study provides a comparative assessment of native freshwater fish biodiversity and identifies the extent to which patterns of taxonomic, functional, and phylogenetic diversity are concordant or divergent across environmental gradients.

The broader relevance of this study lies in clarifying how different facets of biodiversity correspond to one another across large environmental gradients. Rather than assuming that taxonomic, functional, and phylogenetic diversity necessarily show concordant patterns, we empirically compare their spatial distributions and environmental associations within a common continental-scale framework. If strong concordance is observed, this would suggest that taxonomic diversity captures spatial patterns that are broadly shared with functional and phylogenetic diversity. Conversely, divergence among dimensions would indicate that functional and phylogenetic diversity exhibit spatial or environmental patterns that are not fully reflected by species richness. Thus, the results provide empirical evidence on the extent of concordance and divergence among biodiversity dimensions in U.S. freshwater fish assemblages and contribute to a more nuanced understanding of multidimensional biodiversity patterns.

Materials and methods

Freshwater fish occurrence data

This study utilized the most comprehensive watershed-scale (HUC8) database of native freshwater fish distributions in the U.S., compiled by Su et al. (2023) using species-occurrence records from NatureServe (https://www.natureserve.org). This database represents the historical natural distribution of native freshwater fishes across the U.S.. The dataset encompassed 786 freshwater-fish species distributed across 1873 watersheds, providing a robust spatial framework for analyzing and mapping α-diversity patterns in our investigation. This carefully curated database ensured a reliable representation of native freshwater fish distributions across the continent.

Functional traits and phylogenies

We compiled a comprehensive functional-trait dataset for the U.S. freshwater fishes based on the database established by Su et al. (2023), encompassing 20 morphological, physiological and life-history traits. Trait information was sourced from three authoritative databases: the FISHMORPH database (Brosse et al. 2021), the Fish Traits database for U.S. freshwater fishes (Frimpong and Angermeier 2009), and FishBase (Froese and Pauly 2018). These traits were systematically classified into two functional categories: (1) ten traits related to locomotion and feeding (maximum body length, body elongation, relative eye size, oral gape position, relative maxillary length, vertical eye position, body lateral shape, pectoral-fin vertical position, pectoral-fin size, and cauda-peduncle throttling), and (2) ten life-history and physiological traits (longevity, fecundity, age at maturity, trophic level, thermal range, minimum temperature tolerance, maximum temperature tolerance, euryhalinity, parental care, and diet breadth). The selected traits are ecologically meaningful and have been widely used in functional-diversity studies (Brosse et al. 2021), with demonstrated sensitivity to environmental gradients (Villéger et al. 2017). Because functional trait information was missing for some species, Su et al. (2023) used the machine-learning algorithm missForest to impute missing values in the original trait dataset while

incorporating phylogenetic information to improve imputation accuracy. The imputation procedure was extensively validated through repeated simulations and shown to be highly reliable. Therefore, all subsequent analyses in this study were based on the quality-controlled functional trait dataset with imputed missing values provided by Su et al. (2023).

The dataset includes categorical, ordinal, and continuous trait variables, enabling a robust characterization of freshwater fish functional attributes. Pairwise functional dissimilarities among species were calculated using Gower distance (Gower and Legendre 1986), and the resulting distance matrix was subjected to Principal Coordinates Analysis (PCoA). Functional-space quality was evaluated using the mean absolute deviation (MAD) criterion implemented in the R package "mFD" (Maire et al. 2015). Given that improvements in functional-space quality became marginal beyond the fourth axis, the first four principal-coordinate axes were retained for subsequent analyses (Fig. S1). This dimensionality reduction provided a good representation of the original functional distances while minimizing complexity, and the four retained axes collectively explained more than 70% of the total functional variation. Since all watersheds met the criterion of sufficient species richness (≥ 5 species), no watersheds were excluded from the analytical dataset.

We derived phylogenetic distances for 31,526 marine and freshwater ray-finned fish species from the comprehensive phylogeny published by Rabosky et al. (2018). The tree incorporates precise genetic data for 11,638 species, while the remaining 19,888 species were incorporated through stochastic polytomy resolution—a robust phylogenetic imputation method detailed in the original study. This approach maintains evolutionary relationships while maximizing taxonomic coverage, providing a complete phylogenetic framework for our analyses. The resulting phylogeny captures the evolutionary history of both freshwater and marine ray-finned fishes across the U.S., enabling meaningful phylogenetic-diversity comparisons at the watershed scale. Phylogenetic richness, defined as the sum of all branch lengths linking the species in a given assemblage on the phylogenetic tree, served as a measure of phylogenetic diversity (Guégan et al. 1998).

Environmental variables

To assess the environmental associations of freshwater-fish α-diversity patterns, we selected 22 ecologically relevant predictor variables spanning four key environmental dimensions (Table S1): (1) climate anomalies, (2) contemporary temperature, (3) contemporary precipitation, and (4) geographical factors. These predictors were chosen to capture the primary environmental gradients known to influence fish community assembly and biodiversity patterns across U.S. watersheds.

Climate anomalies were the changes in mean annual temperature and precipitation between present and Last Glacial Maximum Temperature (LGMT) and Last Glacial Maximum Precipitation (LGMP), Late Quaternary glacial-interglacial climate-change velocity (Velocity).

Contemporary temperature included annual mean temperature (T), temperature seasonality (TSeasonality), mean temperature of wettest quarter (TWettest), mean temperature of driest quarter (TDriest), mean temperature of warmest quarter (TWarmest), mean temperature of coldest quarter (TColdest), temperature annual range (TAnnualRange), maximum temperature of warmest month (MaxTWarmestMonth), and minimum temperature of coldest month (MinTColdestMonth). "Quarter" denotes a climatological 3-month period.

Contemporary precipitation included annual precipitation (P), precipitation seasonality (PSeasonality), precipitation of wettest quarter (PWettest), precipitation of driest quarter (PDriest), precipitation of warmest quarter (PWarmest), and precipitation of coldest quarter (PColdest). "Quarter" also denotes a climatological 3-month period.

Geographical factors included net primary productivity (NPP), watershed area (Area), slope, and elevation.

The 22 predictor variables were selected based on their established ecological relevance. Collectively capturing climatic variability, current temperature-precipitation regimes, and fundamental landscape characteristics, this comprehensive set of factors allows for a robust evaluation of the environmental factors shaping spatial patterns of α-diversity in U.S. freshwater-fish communities.

Statistical analysis

All statistical analyses and data visualization processes were performed using R software version 4.4 (R Core Team 2024). We systematically evaluated multiple dimensions of freshwater-fish α-diversity. TD was quantified as species richness based on species-occurrence data. Functional diversity was quantified for each watershed using the TPD framework via the accompanying "TPD" R package (Carmona 2019). In this study, FD was represented by functional richness, which characterizes the volume of trait space occupied by the species pool within each respective watershed. Phylogenetic diversity was assessed using Faith's phylogenetic-diversity index, computed with the "pd" function from the "picante" package (Kembel et al. 2010).

To reduce the direct influence of species richness on functional and phylogenetic diversity, we applied a null-model standardization approach. Specifically, the Curveball algorithm was used to generate 999 randomized matrices by reshuffling the community matrix while preserving both the species richness of each watershed and the occurrence frequency of each species, that is, the row and column marginal totals of the community matrix (Strona et al. 2014). Standardized effect sizes (SES) were then calculated by comparing the observed FD and PD with their corresponding null distributions, i.e., as the difference between observed and expected values divided by the standard deviation of the null distribution. Thus, sesFD and sesPD quantify the extent to which the observed functional and phylogenetic diversity deviates from the expectation under a null model conditional on the observed species richness and species occurrence frequencies. This standardization was intended to reduce the direct influence of species richness on FD and PD, rather than to assume complete statistical independence between FD or PD and TD. The ecological interpretation of these SES values is therefore conditional on the constraints imposed by the Curveball null model. In particular, positive or negative SES values indicate departures from the functional or phylogenetic diversity expected under randomized assemblages that retain the observed species richness of each watershed and the observed occurrence frequency of each species. Positive SES values indicate higher FD or PD than expected by chance (i.e., functional or phylogenetic overdispersion), whereas negative values indicate lower values than expected (i.e., functional or phylogenetic clustering). Hereafter, when mentioning FD and PD, we always refer to sesFD and sesPD, which were used in the subsequent analyses. To assess the degree of concordance among these biodiversity facets, we performed pairwise Spearman rank correlation analyses among TD, sesFD, and sesPD diversity metrics. This integrated analytical approach not only characterizes the multifaceted nature of α-diversity but also reveals potential synergies or divergences among different aspects of community organization.

We employed Boosted Regression Trees (BRTs) to quantify the relative influence of environmental predictors on TD, sesFD, and sesPD diversity patterns (Elith et al. 2008). Although BRT models are inherently robust to collinearity among predictors, we adopted a conservative approach by removing 12 variables based on their environmental categories and correlation strengths. Following Parravicini et al. (2013), variables were excluded if they exhibited strong pairwise correlations (absolute Spearman's $R > 0.7$) or represented redundant ecological information. The final BRT models incorporated 10 carefully selected predictors from the original 22-variable set (Fig. S2), ensuring both model robustness and ecological interpretability. This rigorous variable-selection process not only mitigated potential multicollinearity effects but also enhanced our capacity to discern the distinct contributions of key environmental variables to each biodiversity dimension.

We implemented BRT modeling following the established framework of Elith et al. (2008). The BRT approach incorporates four key hyperparameters: (1) learning rate (lr) controlling individual tree contributions, (2) bag fraction (bf) determining subsampling proportion, (3) tree complexity (tc) governing interaction depth, and (4) number of trees (nt) defining model iterations. To optimize model performance, we conducted a rigorous two-stage calibration process. First, we performed 10-fold cross-validation to identify parameter combinations maximizing cross-validated deviance explained ($CV\text{-}D^2$). Subsequently, we enhanced model stability through 100 bootstrap iterations with randomized seeds (Su et al. 2023), from which we derived mean estimates of predictor importance and explained deviance (D^2). We further examined bivariate predictor interactions using the "gbm.interactions" function in the "dismo" package

(Elith et al. 2008). This comprehensive analytical framework ensures robust model specification and provides detailed insights into: (1) the relative influence of environmental predictors, (2) their nonlinear effects, and (3) potential interactions shaping biodiversity patterns.

Considering the methodological limitations of BRT models in handling spatial autocorrelation, we additionally applied a spatial autoregressive error (SARerror) model to reanalyze the three α-diversity indices, in order to evaluate whether spatial effects influenced the inferences drawn from the BRT results. All predictor variables were standardized prior to analysis (mean = 0, variance = 1) to ensure comparability among predictors. The SARerror model incorporated spatial constraints through a spatial weights matrix based on watershed adjacency relationships, constructed using the "poly2nb" and "nb2listw" functions in the "spdep" package. Spatial modeling was implemented within the framework of the "spatialreg" and "spdep" packages (Bivand and Piras 2015). Model performance was assessed using the Nagelkerke pseudo-R^2, and the spatial independence of residuals was evaluated using global Moran's I tests (Cliff and Ord 1981; Nagelkerke 1991). Comparison of results (Tables S2–S4) showed that the key environmental variables identified by the BRT models remained largely consistent in the SARerror models, indicating that our main conclusions are robust and not sensitive to spatial autocorrelation.

Results

Spatial patterns of α-diversity and relationships with environmental variables

Across the 1873 watersheds, the three α-diversity facets exhibited distinct spatial patterns. For TD, most watersheds contained fewer than 60 species, whereas only 11 watersheds exhibited exceptionally high species richness, including 10 watersheds with 120–150 species and one watershed with more than 150 species. Spatially, TD was lowest in the Rocky Mountains and the western U.S., increased eastward across the Central Plains, and then declined again in the Appalachian Mountains and adjacent regions of the eastern U.S. (Fig. 1a). The spatial pattern of sesFD differed markedly from that of TD. Overall, 1467 watersheds (78%) exhibited negative SES values for FD, indicating that observed FD was lower than expected under random community assembly given the observed species richness. In contrast, only 406 watersheds showed positive FD values, indicating higher FD than expected by chance (Fig. 1b). sesFD hotspots were primarily distributed along the Pacific Coast, the Atlantic Coast, and the Florida Peninsula, whereas most watersheds in the Central Plains exhibited negative FD values. The spatial distribution of sesPD partially overlapped with that of sesFD but differed substantially from that of TD (Fig. 1c). High SES values for PD were concentrated along the Atlantic Coast, particularly around the Appalachian Mountains, as well as in the Florida Peninsula and the Great Lakes region, indicating that species within these communities were more distantly related than expected under the null model given the observed species richness. In contrast, most watersheds in the western U.S., especially the Rocky Mountains, together with several watersheds in the Central Plains, exhibited low sesPD values, suggesting stronger phylogenetic clustering than expected by chance, potentially reflecting habitat filtering that preferentially selects species from specific evolutionary lineages. Although sesPD and sesFD shared hotspot regions along the Atlantic Coast and the Florida Peninsula, sesPD showed a distinct hotspot in the Great Lakes region, whereas sesFD reached its highest values along the Pacific Coast.

The relative influence of environmental variables varied substantially among the three biodiversity facets. Environmental variables explained 86.9% of the variation in TD, with PWarmest emerging as the dominant predictor (49.1%) (Fig. 1d). sesFD was moderately well explained (65.5%), primarily by elevation (31.6%), followed by LGMT (14.9%) (Fig. 1e). By comparison, sesPD showed considerably lower explanatory power (53.1%), with elevation accounting for more than one-third (35.3%) of the explained variation, followed by LGMT (14.6%), while the contributions of all remaining predictors were relatively small (Fig. 1f). Despite these differences, precipitation, elevation and LGMT were consistently identified as important predictors across all three biodiversity facets, although their relative influence varied markedly among them.

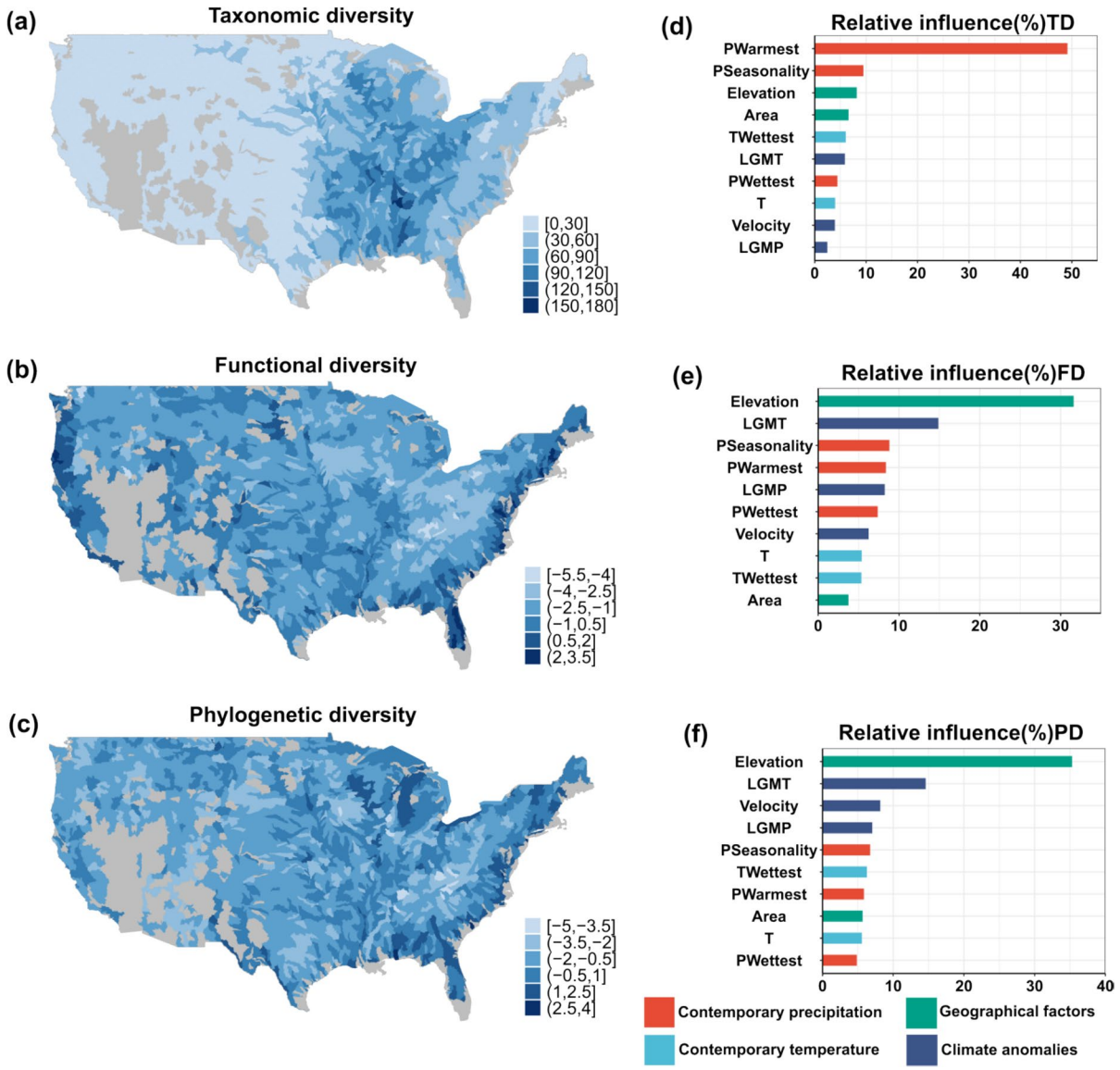


Fig. 1 Alpha diversity and environmental variables for freshwater fishes in the U.S. watersheds. **a** Map of taxonomic diversity (TD); **b** map of functional diversity (sesFD); **c** map of phylogenetic diversity (sesPD); **d** relative influence of 10 factors affecting taxonomic diversity; **e** relative influence of 10 factors affecting functional diversity; **f** relative influence of 10 factors affecting phylogenetic diversity

Relationship of the three facets of α-diversity

Spearman correlation analysis revealed contrasting relationships among the three facets of α-diversity across the 1873 watersheds (Fig. 2). TD was negatively correlated with sesFD ($R = -0.19$, $P < 4.1 \times 10^{-16}$), whereas sesPD was positively correlated with both TD ($R = 0.22$, $P < 2.2 \times 10^{-16}$) and sesFD ($R = 0.58$, $P < 2.2 \times 10^{-16}$). Because sesFD and sesPD were standardized against a null model that preserves both watershed-level species richness and species occurrence frequencies, these correlations should be interpreted as associations among richness-standardized diversity metrics rather than as evidence that sesFD and sesPD are statistically independent of TD. Overall, the

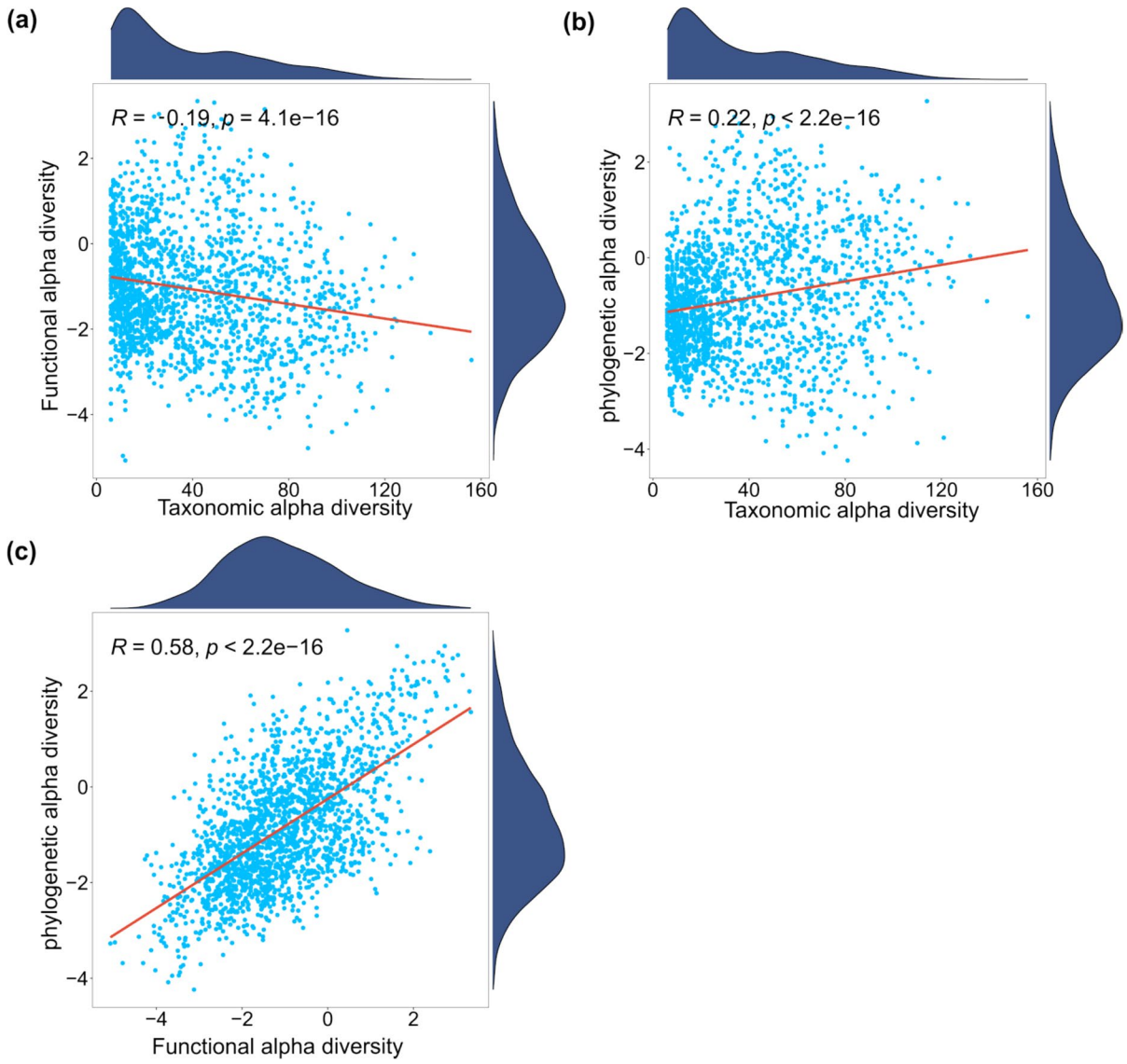


Fig. 2 Correlation plots of α-diversity phenomena for freshwater fishes across the U.S. watersheds. **a** Correlation plot of taxonomic and functional diversity (sesFD); **b** correlation plot of taxonomic and phylogenetic diversity (sesPD); **c** correlation plots of functional and phylogenetic diversity. Correlation coefficients (R) are shown for each relationship based on Spearman test ($p < 0.05$)

strongest association was observed between sesFD and sesPD, whereas the relationship between TD and sesPD was comparatively weak. TD and sesFD exhibited a significant negative correlation. These contrasting correlations indicate that TD, sesFD, and sesPD were not consistently associated across watersheds, suggesting that the three facets capture partly distinct dimensions of α-diversity.

Nonlinear responses and interactions among environmental variables

The BRT models explained substantial proportions of variation across all three α-diversity dimensions, although the explanatory power varied markedly among them (Fig. 3). For TD, the model performed best, with PWarmest emerging as the

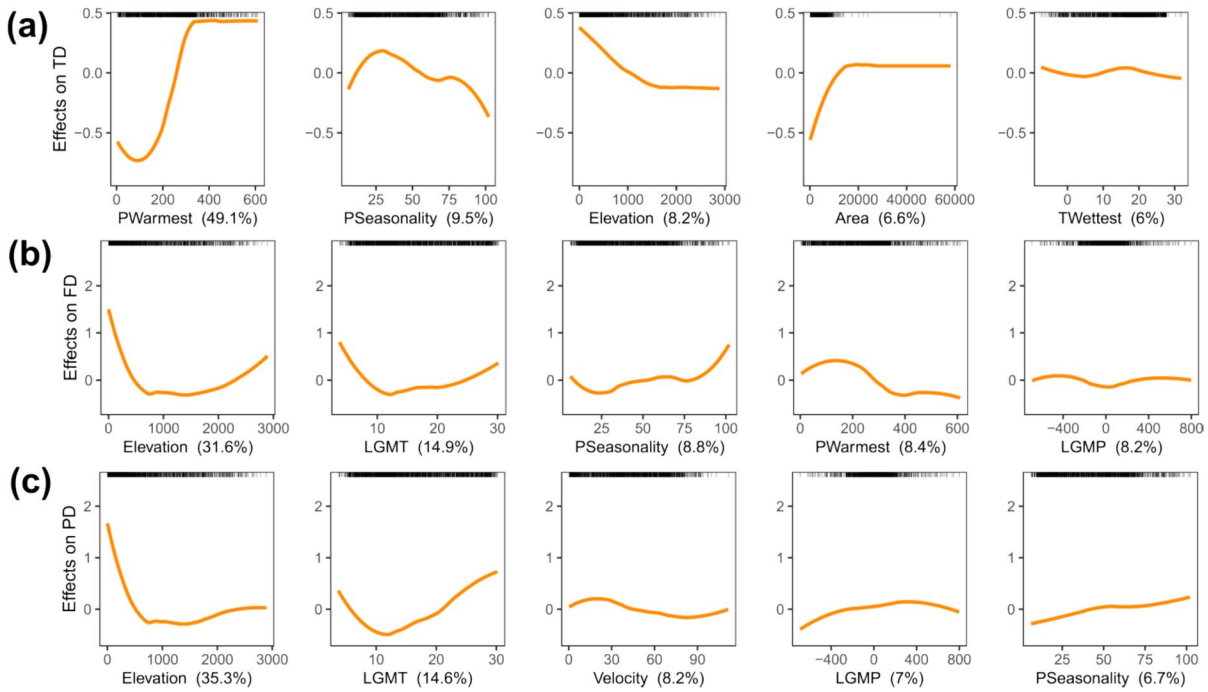


Fig. 3 Results of boosted regression trees showing the partial dependency between three facets of α-diversity and the first five predictors. The value in parentheses in each panel shows the percentage of contribution of each predictor considered in the model. The rugs in the top of each panel show the distribution of the watersheds along the predictor values. PWarmest: precipitation of warmest quarter; PSeasonality: precipitation seasonality; Area: watershed area; TWettest: mean temperature of wettest quarter; LGMT: change in mean annual temperature between present and Last Glacial Maximum conditions; LGMP: changes in mean annual precipitation between present and Last Glacial Maximum Precipitation; Velocity: Late Quaternary glacial-interglacial climate-change velocity

dominant predictor. TD exhibited a pronounced nonlinear response, increasing rapidly beyond an intermediate threshold of PWarmest and eventually reaching a plateau, indicative of a typical saturating response (Fig. 3a). The environmental predictors of sesFD exhibited a more complex structure. It was primarily controlled by elevation, with additional contributions from LGMT. Overall, sesFD increased nonlinearly with elevation, with the lowest diversity observed at intermediate elevations and higher values at both lower and higher elevations. Similarly, the effect of LGMT on sesFD followed a pattern comparable to that of elevation (Fig. 3b). In comparison, sesPD exhibited weaker environmental responsiveness but still showed clear structured patterns. Elevation emerged as the dominant factor, exerting a distinct threshold effect, whereby sesPD declined sharply at low elevations and then stabilized at higher elevations. LGMT also showed a nonlinear effect on sesPD, with an initial decrease followed by an increase, suggesting complex ecological filtering processes (Fig. 3c). Overall, the responses of the three α-diversity dimensions to environmental gradients were predominantly nonlinear, although the key predictors differed substantially among dimensions. TD was mainly constrained by precipitation- and elevation-related factors, whereas both sesFD and sesPD were primarily shaped by topographic gradients and LGMT. Although with comparatively lower relative contributions, the remaining environmental predictors still displayed clear and ecologically meaningful response patterns, as illustrated in Figs. S3–S5.

Significant interactions among environmental variables were detected across all three diversity dimensions (Fig. 4). For TD, PWarmest showed the strongest interactions with PSeasonality and LGMT, indicating that the effects of warm-season precipitation were jointly modulated by climatic variability and historical climate conditions. For sesFD, the strongest interactive effect was detected between

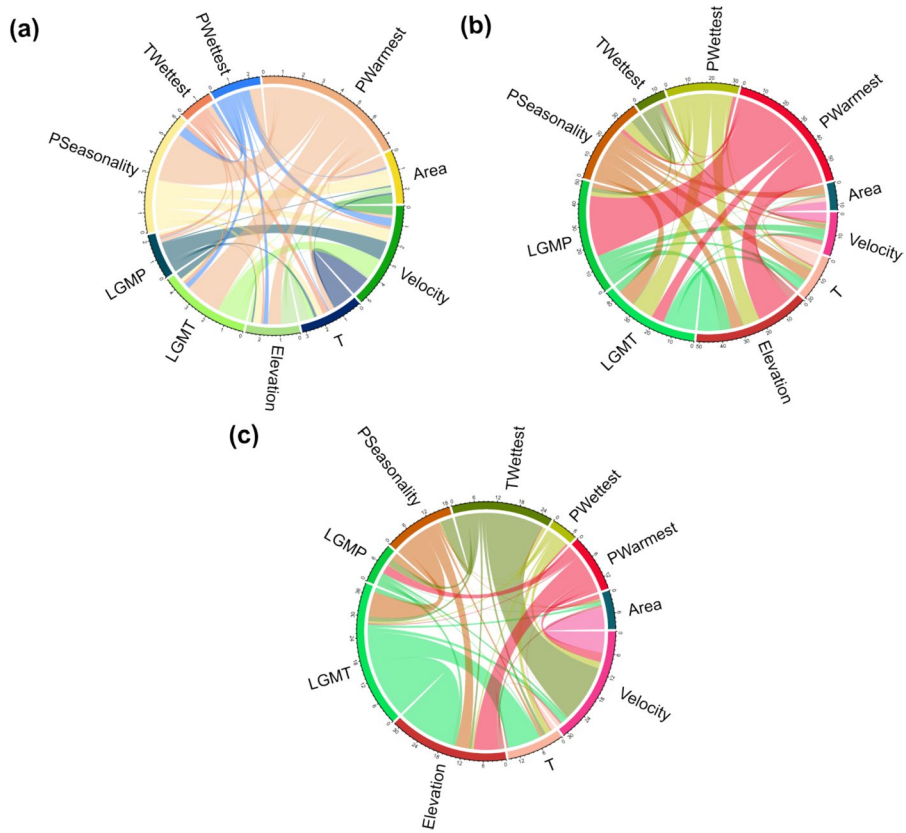


Fig. 4 Interactions between the 10 predictors for each boosted regression trees (BRT) model. The width of the bands represents the strength of interaction between pairs of the predictors used to build the BRT models for **a** taxonomic diversity, **b** functional diversity (sesFD) and **c** phylogenetic diversity (sesPD). Area: watershed area; Velocity: Late Quaternary glacial-interglacial climate-change velocity; T: annual mean temperature; LGMT & LGMP: change in mean annual temperature and precipitation between present and Last Glacial Maximum conditions; PSeasonality: precipitation seasonality; TWettest: mean temperature of wettest quarter; PWettest: precipitation of wettest quarter; PWarmest: precipitation of warmest quarter

PWarmest and the combined factors of elevation and LGMP, indicating that the influence of warm-season precipitation on sesFD was strongly mediated by both topographic conditions and glacial-interglacial precipitation anomalies. For sesPD, the dominant interaction was observed between LGMT and elevation, with a secondary interaction between TWettest and velocity, suggesting that sesPD was jointly shaped by long-term climatic history, topographic gradients, and the legacy of past climate dynamics.

Discussion

Our results show that TD, sesFD and sesPD are not interchangeable descriptors of freshwater fish biodiversity. Although they exhibited partial spatial congruence, each diversity dimension displayed distinct environmental associations, reflecting different ecological and evolutionary processes. These findings underscore the value of integrating multiple biodiversity dimensions to better understand large-scale biodiversity patterns and the factors that shape them (Strecker et al. 2011; Liu et al. 2023).

Patterns and relationships among the three facets of α-diversity

Our study revealed pronounced spatial heterogeneity in the taxonomic, functional, and phylogenetic dimensions of α-diversity for native freshwater fishes across the United States, demonstrating that these three facets represent distinct components of biodiversity organization. Although all three dimensions were influenced by shared environmental gradients, their spatial patterns and key environmental correlates differed substantially, suggesting that species richness, ecological functions, and evolutionary history may be shaped by partly different processes during community assembly, although these mechanisms were not directly tested in this study.

TD exhibited a pronounced west–east gradient, with relatively low values in the mountainous western U.S. and higher values across the Central

Plains and eastern regions. This pattern is closely associated with large-scale topographic and climatic gradients across the continent. The western U.S. is characterized by extensive mountain ranges and plateaus with high elevations and relatively arid climates, whereas the eastern U.S. has lower elevations, greater precipitation, and more extensive freshwater drainage networks (Hamilton and Myers 1966). The BRT analysis further identified precipitation during the warmest quarter as the primary predictor of taxonomic diversity. Its nonlinear association suggests that water availability is closely related to broad-scale variation in freshwater fish species richness. One possible explanation is that precipitation affects hydrological conditions and habitat availability, which may in turn influence opportunities for dispersal and colonization. In high-elevation regions of the western U.S., lower taxonomic diversity may plausibly arise from geographic isolation, reduced river connectivity, and limited dispersal opportunities (Jaramillo-Villa et al. 2010; Bhatt et al. 2013). However, this interpretation remains a hypothesis rather than a demonstrated conclusion.

The spatial pattern of sesFD differed markedly from that of TD. High sesFD values were primarily concentrated along the Pacific Coast, the eastern Appalachian region, and the Florida Peninsula, whereas the Central Plains generally exhibited low sesFD. Consistent with the BRT results, elevation emerged as one of the most important predictors, suggesting that topographic gradients and the associated environmental variation are strongly associated with the functional structure of freshwater fish communities (Troia and Gido 2015; Kirk et al. 2022). The Central Plains are characterized by relatively flat terrain and environmentally homogeneous conditions. This setting may offer limited opportunities for niche differentiation and could promote the coexistence of species with similar functional traits, consistent with the observed FD lower than expected under random assembly (Bower and Winemiller 2019). In contrast, regions such as the Pacific Coast and the eastern Appalachian Mountains are characterized by complex topography, diverse hydrological conditions, and heterogeneous habitat types. This environmental heterogeneity may increase variation in resource availability and local environmental conditions, potentially facilitating niche differentiation and allowing communities to occupy a larger functional trait space (Agra et al. 2024).

The spatial pattern of sesPD partially overlapped with that of sesFD but was not entirely congruent. High sesPD values were primarily concentrated along the Atlantic Coast, the Florida Peninsula, and the Great Lakes region, whereas relatively low values were observed throughout the mountainous western U.S., particularly in the Rocky Mountains. The BRT analysis showed that elevation was the strongest predictor of sesPD, revealing a pronounced threshold response, suggesting that topographic gradients and their associated hydrological and environmental conditions play a critical role in shaping the phylogenetic composition of freshwater fish communities. In high-elevation regions of the western U.S., complex topographic barriers, reduced watershed connectivity, and the limited availability of suitable habitats may restrict lineage dispersal and regional species exchange. If so, communities would tend to be dominated by a relatively small number of environmentally adapted lineages, which is consistent with the lower PD observed in these regions (Xia et al. 2024; Mao et al. 2026).

Our analysis revealed pronounced incongruence among the three α-diversity facets of native freshwater fishes in the U.S., indicating that taxonomic, functional, and phylogenetic diversity capture different aspects of community organization processes (Liang et al. 2025). TD was significantly and positively correlated with sesPD, although the strength of this relationship was relatively weak, suggesting that increases in species richness do not necessarily result in proportional increases in richness-standardized phylogenetic diversity (Knouft and Page 2011). This finding is consistent with Du et al. (2026), who reported only moderate concordance between species richness and PD in schizothoracine fishes on the Qinghai–Tibet Plateau. High species richness may partly result from the diversification or expansion of closely related lineages, leading to increases in TD without equivalent gains in PD (Du et al. 2026). In contrast, TD showed a significant negative relationship with sesFD, indicating that increases in species richness did not necessarily correspond to an expansion of functional space beyond that expected under the null model. This pattern may result from the combined effects of environmental filtering and niche selection (Liang et al. 2025). Environmental filtering

can favor the coexistence of species with similar functional traits under specific environmental conditions, potentially resulting in functional clustering and increased overlap in functional strategies among coexisting species (HilleRisLambers et al. 2012). Under this hypothesized mechanism, regions with higher species richness may be dominated by strong environmental constraints, causing coexisting species to converge toward similar ecological strategies and limiting the expansion of functional trait space. Conversely, species-poor regions with greater environmental heterogeneity or stronger niche differentiation may maintain higher functional divergence. sesFD exhibited a strong positive relationship with sesPD, suggesting that evolutionary history is strongly associated with the richness-standardized functional structure of freshwater fish communities. Long-term lineage diversification may facilitate the accumulation of divergent ecological strategies and functional traits, resulting in communities with greater phylogenetic separation also tending to possess higher functional diversity. However, the strength of phylogenetic constraints varies substantially among different functional traits (Cano-Barbacil et al. 2022; Liang et al. 2024). For example, Ster; Liang et al. 2024). For example, Sternberg and Kennard (2014) demonstrated that relationships between functional traits and phylogeny varied among Australian freshwater fishes, with some traits exhibiting strong phylogenetic conservatism, whereas others were relatively decoupled from evolutionary history. Such trait-specific variation may explain why, despite the overall strong association between sesFD and sesPD observed in our study, some functional attributes showed weaker relationships with phylogenetic structure.

Relationships between environmental factors and α-diversity patterns

Identifying the environmental factors associated with biodiversity variation is essential for biodiversity conservation and the development of effective management strategies (He et al. 2020; Su et al. 2021). Our study identified contemporary precipitation regimes, elevation, and LGMT as the environmental variables most strongly associated with multidimensional biodiversity patterns.

Our results indicate that precipitation is particularly strongly associated with TD. During the warmest period, increased precipitation is expected to promote flooding and enhance habitat connectivity by creating complex aquatic habitat networks (Thomaz et al. 2007), which could in turn facilitate fish dispersal and promote frequent restructuring of fish assemblages among watersheds (Oliveira et al. 2015). Under this hypothesis, precipitation-driven increases in hydrological connectivity would expand available habitats and resource availability, thereby potentially supporting higher species diversity (Fullerton et al. 2010; Garbin et al. 2019; Xiong et al. 2023). Precipitation seasonality exhibited a hump-shaped relationship with TD. Moderate levels of precipitation seasonality may increase habitat heterogeneity and create additional ecological niches, thereby promoting species coexistence. However, excessive precipitation seasonality could intensify hydrological extremes, such as droughts and floods, reducing habitat stability and ultimately decreasing TD. This pattern suggests that intermediate levels of hydrological variability, rather than either highly stable or highly extreme conditions, may be more favorable for maintaining freshwater fish diversity (van der Sleen and Rams 2023).

The nonlinear variation of sesFD along the elevational gradient is consistent with a potential shift in community assembly processes across elevation. From low to intermediate elevations, harsh abiotic conditions may selectively favor species with similar functional traits, resulting in reduced FD (Kirk et al. 2022; Poveda-Cuellar et al. 2022). The increase in sesFD at high elevations does not necessarily indicate a relaxation of environmental constraints; rather, it may reflect increased habitat heterogeneity. Such heterogeneity could provide greater opportunities for niche differentiation among species that have persisted after environmental filtering, potentially facilitating a transition from environmentally filtered assembly toward limiting similarity processes. Similarly, Myers et al. (2021) found that functional α-diversity of marine fishes increased with depth, with both intraspecific and interspecific competition acting as key mechanisms shaping functional multidimensional space. These competitive interactions may promote niche partitioning and facilitate species coexistence. Under such conditions, coexisting species tend to diverge in key traits associated with resource acquisition, particularly those related to trophic strategies (Poveda-Cuellar et al. 2022), thereby reducing

interspecific competition and supporting higher levels of functional diversity at the community level.

sesPD exhibited a relatively consistent negative association with elevation. As elevation increased, PD declined significantly, indicating a strong relationship between elevational gradients and the phylogenetic structure of freshwater fish communities. One possible explanation is that changes in thermal conditions, hydrological characteristics, habitat availability, and watershed connectivity along the elevational gradient may influence the distribution of evolutionary lineages (Xia et al. 2024). At lower elevations, relatively stable thermal and hydrological conditions, together with well-developed river networks, may provide a wider range of environmental conditions and greater opportunities for the occurrence and coexistence of lineages. In addition, higher primary productivity may be associated with greater availability of ecological resources and a broader range of ecological conditions, potentially contributing to the presence of lineages with different ecological characteristics (Xia et al. 2023, 2024). In contrast, high-elevation streams are typically characterized by lower temperatures, stronger hydrological variability, and greater environmental constraints. These conditions may be associated with the lower occurrence of lineages with contrasting environmental requirements, resulting in lower phylogenetic diversity (Bower and Winemiller 2019; Xia et al. 2024). Although PD showed a slight recovery at the highest elevations, the magnitude of this increase was limited, potentially reflecting localized variation in habitat conditions and environmental heterogeneity.

LGMT exhibited significant U-shaped relationships with both sesFD and sesPD. In regions with intermediate LGMT values and relatively low elevations, historical climatic fluctuations combined with environmental filtering may have selectively favored species with similar tolerance ranges, potentially resulting in functional and phylogenetic clustering and consequently lower FD and PD. In contrast, mountainous regions characterized by high LGMT values and high elevations generally possess complex topography and greater habitat heterogeneity, which could support species with diverse ecological strategies and evolutionary backgrounds. For example, in prairie streams, stronger filtering of reproductive strategies occurs in upstream reaches, whereas downstream environments impose greater selection on feeding strategies (Troia and Gido 2015). This transition may drive community structure from clustering toward greater functional and phylogenetic divergence, resulting in increased FD and PD (Schwilk and Ackerly 2005).

Conservation implications

Our findings provide implications for freshwater fish biodiversity assessment and conservation planning in the U.S. The pronounced spatial heterogeneity and partial incongruence among TD, FD, and PD suggest that conservation planning may benefit from considering multiple biodiversity dimensions rather than relying solely on TD (Strecker et al. 2011; Liu et al. 2023). These differences indicate that areas identified as priorities from different biodiversity dimensions may not fully overlap. Watersheds with high TD may be particularly relevant for maintaining species diversity, given that taxonomic diversity reflects the number of coexisting species and is related to population persistence and dispersal opportunities (Perkin et al. 2015). In contrast, watersheds with relatively high FD or PD may contain distinctive functional or evolutionary compositions that are not fully reflected by species richness. Conservation planning could therefore consider TD, FD, and PD together with habitat condition, connectivity, and environmental heterogeneity when characterizing freshwater fish biodiversity. In particular, incorporating geodiversity and habitat heterogeneity may help account for environmental variation associated with freshwater biodiversity patterns (Wang et al. 2025). Our environmental analyses further showed that precipitation, elevational gradients, and historical climatic changes were differently associated with the three biodiversity dimensions. These patterns highlight the potential relevance of both contemporary environmental conditions and historical climatic legacies when assessing regional conservation priorities. For low-elevation river systems, maintaining river connectivity, natural hydrological regimes, and critical habitats may be important for conserving freshwater fish assemblages (He et al. 2017; Stoffers et al. 2022). Meanwhile, some mountainous watersheds with relatively low TD may possess high functional or phylogenetic uniqueness.

Limitations

Although this study provides a multidimensional assessment of freshwater fish diversity across the U.S., several limitations should be acknowledged. First, species distribution data were primarily based on historical occurrence records and may not fully capture recent distributional changes caused by human activities and climate change. Second, although the available functional trait dataset includes morphological, life-history, and ecological characteristics, information on dispersal ability and other relevant traits remains limited, which may constrain a comprehensive representation of functional space. Third, some species in the phylogenetic tree were supplemented based on information from closely related taxa, which may introduce a degree of phylogenetic uncertainty. Future studies incorporating higher spatiotemporal resolution monitoring data, more comprehensive functional trait information, and process-based ecological models will help further validate and advance the ecological interpretations proposed in this study. An additional limitation concerns the interpretation of environmental associations. Our analyses identify statistical relationships between diversity metrics and broad-scale environmental variables but do not directly evaluate the community-assembly mechanisms that may contribute to these patterns. Therefore, interpretations involving environmental filtering, limiting similarity, niche differentiation, dispersal limitation, or habitat heterogeneity should be viewed as plausible explanations rather than mechanisms directly demonstrated by the present study. Future research integrating higher-resolution spatiotemporal monitoring, more comprehensive functional trait data, mechanistic community-assembly models, demographic information, and future climate projections will be important for testing these hypotheses.

Conclusion

This study examined the spatial patterns and environmental associations of taxonomic, functional, and phylogenetic α-diversity across 1,873 watersheds containing native freshwater fish species in the contiguous U.S. The three diversity facets showed pronounced spatial heterogeneity and only partial spatial congruence. TD generally increased from west to east, whereas sesFD and sesPD exhibited distinct hotspot distributions. Their relatively weak correlations further indicated that spatial patterns of the three facets were not fully concordant. Environmental associations also differed among facets: precipitation during the warmest quarter was most strongly associated with TD, whereas elevation and LGMT were more important for sesFD and sesPD. These results indicate that contemporary environmental gradients and historical climatic conditions are associated with different facets of freshwater fish α-diversity. Overall, our study provides a continental-scale comparison of taxonomic, functional, and phylogenetic diversity and highlights both shared and distinct biodiversity patterns across U.S. freshwater fish assemblages. Considering multiple biodiversity facets may therefore provide a more nuanced basis for freshwater fish biodiversity assessment and conservation planning. Future studies incorporating higher-resolution data and mechanistic approaches are needed to further evaluate the processes underlying these patterns.

Author contributions C.X., N.L., J.X., and G.S.: Conceptualization; C.X., C.D., N.L., and G.S.: Methodology; C.X., Z.C., J.L., Y.L., C.D., and G.S.: Formal analysis; C. X.: Writing—Original Draft and Visualization; G.S.: Data Curation; J.X., and G.S.: Supervision; J.L. and G.S.: Funding acquisition; All authors: Writing—Review & Editing.

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Data availability The data supporting the findings of this study are available from [Zenodo] at https://doi.org/10.5281/zenodo.21494791 and our Supplementary Information.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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